

Chess Game - Multiplayer Edition

A fully-featured, browser-based chess game with complete rule implementation, light/dark mode support, and comprehensive game statistics tracking.

Features

Complete Chess Rules Implementation

-  All standard piece movements (Pawns, Rooks, Knights, Bishops, Queens, Kings)
-  Special moves:
 - **Castling** (both kingside and queenside)
 - **En Passant** capture
 - **Pawn Promotion** (with piece selection dialog)
-  Game state detection:
 - **Check** detection and warning
 - **Checkmate** detection
 - **Stalemate** detection
 - **50-move rule** for draws
-  Move validation (prevents illegal moves and self-check)
-  Visual indicators for valid moves and captures

Multiplayer Support

-  Turn-based local multiplayer (two players on the same device)
-  Turn indicator showing current player
-  Move history tracking with algebraic notation
-  Captured pieces display for both players

Light & Dark Mode

-  Toggle between light and dark themes
-  Theme preference saved to browser localStorage
-  Smooth theme transitions
-  Optimized colors for both modes

Game Statistics Dashboard

-  Track all game statistics:
 - Total games played

- White wins
- Black wins
- Draws
- Checkmates
- Stalemates
- 📅 Recent game history (last 20 games)
- 📅 Game details: date, moves, result type
- 📅 Persistent storage using localStorage

Modern UI/UX

- 🎨 Clean, responsive design
- 🎨 Visual feedback for piece selection
- 🎨 Highlighted valid moves and capture opportunities
- 🎨 Mobile-friendly layout
- 🎨 Smooth animations and transitions



Getting Started

Prerequisites

- A modern web browser (Chrome, Firefox, Safari, Edge)
- No server installation required!

Installation & Running

1. Clone the repository:

```
git clone https://github.com/AyshaAnsari/Chess.git
cd Chess
```

2. Open the game: Simply open index.html in your web browser, or serve it using a local server:

```
# Using Python 3
python3 -m http.server 8000
```

```
# Using Node.js (if you have http-server installed)
npx http-server
```

3. Play: Navigate to http://localhost:8000 in your browser.



How to Play

1. Starting a Game:

- The board is set up automatically with white pieces at the bottom
- White always moves first

2. Making Moves:

- Click on a piece to select it
- Valid moves will be highlighted (dots for empty squares, red border for captures)
- Click on a highlighted square to move the piece
- Click elsewhere to deselect

3. Special Moves:

- **Castling:** Select your king and click two squares toward the rook
- **En Passant:** Available when opponent moves pawn two squares
- **Pawn Promotion:** A dialog appears when a pawn reaches the opposite end

4. Game Controls:

- 🌓 **Toggle Theme:** Switch between light and dark modes
- 🎮 **New Game:** Start a fresh game
- 📊 **Dashboard:** View game statistics and history

5. Winning the Game:

- Checkmate your opponent's king to win
- The game also ends in a draw (stalemate, 50-move rule)

Project Structure

Chess/

— index.html	# Main HTML structure
— style.css	# Styling with light/dark mode
— chess.js	# Chess game logic and rules
— app.js	# UI and application logic
— README.md	# This file



Technical Details

Technologies Used

- **HTML5:** Semantic markup
- **CSS3:** Modern styling with CSS variables for theming
- **JavaScript (ES6+):** Game logic and DOM manipulation
- **LocalStorage API:** Persistent data storage

Browser Compatibility

- Chrome/Edge 90+
- Firefox 88+
- Safari 14+
- Opera 76+

Customization

The game uses CSS variables for theming. You can easily customize colors by modifying the `:root` and `[data-theme="dark"]` sections in `style.css`:

```
:root {  
  --bg-primary: #f0f0f0;  
  --square-light: #f0d9b5;  
  --square-dark: #b58863;  
  /* ... more variables */  
}
```

Contributing

Contributions are welcome! Feel free to:

- Report bugs
- Suggest new features
- Submit pull requests

License

This project is open source and available for educational purposes.

Game Rules Reference

For those new to chess, here are the basic rules:

- **Pawns:** Move forward one square (two on first move), capture diagonally
- **Rooks:** Move horizontally or vertically any number of squares
- **Knights:** Move in an L-shape (2+1 squares), can jump over pieces
- **Bishops:** Move diagonally any number of squares
- **Queen:** Combines rook and bishop movements
- **King:** Moves one square in any direction

The goal is to checkmate the opponent’s king (put it under attack with no legal escape).

Acknowledgments

Created with ♥ for chess enthusiasts and developers learning game development!

Enjoy playing chess!      